Synthesis of Pentyl Valerate via Fischer Esterification

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1 Introduction

The objective of this experiment is to synthesize pentyl valerate via Fischer Esterification by reacting 1-hexanol with valeric acid in the presence of catalytic concentrated sulfuric acid. The product will be characterized and analyzed by odor analysis, ATR-IR, 100 MHz ^{1}H NMR, and GC-MS.

2 Reaction Scheme

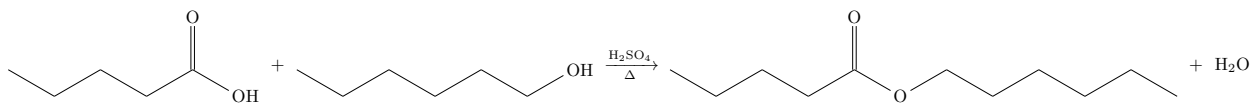


Figure 1: Reaction of Valeric Acid and 1-Pentanol

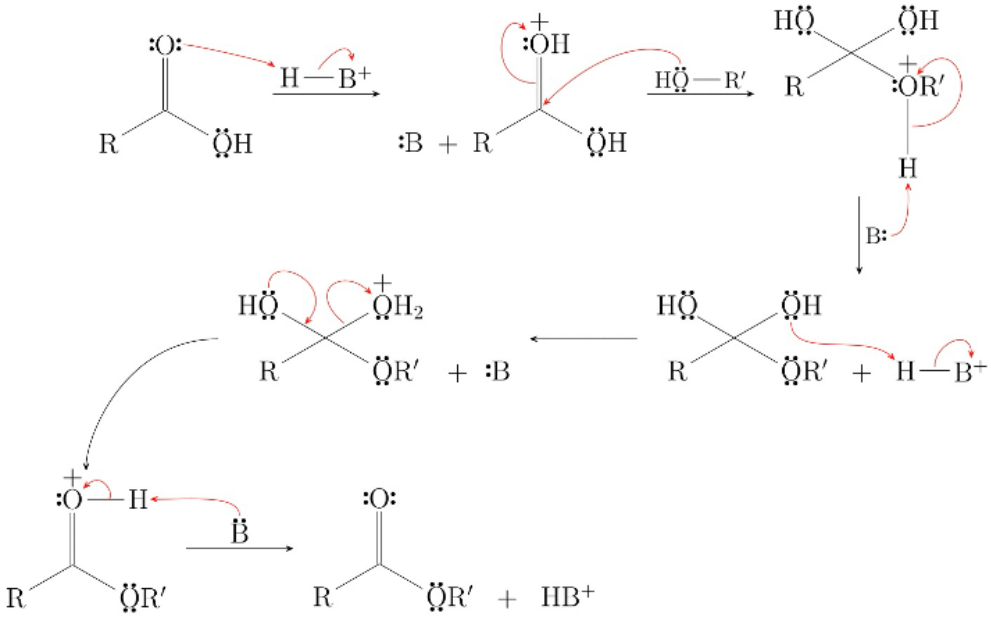


Figure 2: Fischer Esterification Mechanism

3 Table of Reagents and Products

Compound	MW (g/mol)	Density (g/mL)	B.P (°C)	Equiv	Amount	mmol
Valeric Acid	102.13	0.930 - 0.939	185 - 186	1.0	4.77 g / 5.07 mL	46.7
1-Hexanol	102.18	0.814	155	3.0	9.54 g / 11.63 mL	93.4
Sulfuric Acid	98.08	1.84	–	cat.	3 drops	–
Diethyl Ether	74.12	0.710	34.6	solv.	–	–
Hexyl Valerate	186.29	0.867	224 - 226	–	–	–

4 Experimental Procedure

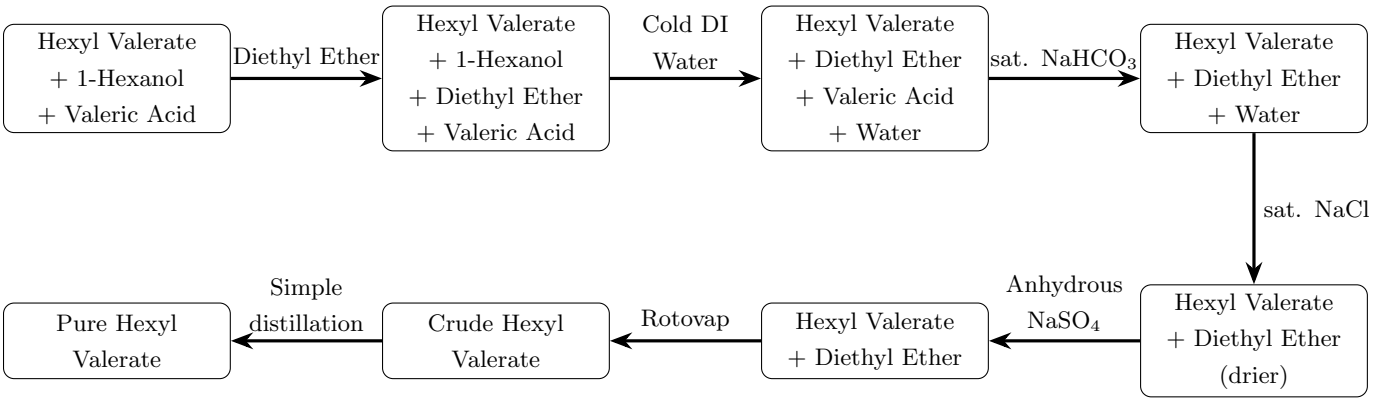
4.1 Apparatus

- Reflux
- Simple Distillation
- Rotovap

4.2 Procedure

- In a 25 mL round-bottom flask, add 4.77g (5.07 mL) of valeric acid followed by 9.54 g (11.63 mL) of 1-pentanol and a stir bar.
- While stirring, slowly add 3 drops of concentrated sulfuric acid.
- Gently reflux for 1 hour, then allow the mixture to cool to room temperature.
- Chill the product in an ice bath then transfer the mixture to a separatory funnel using 20 mL total of diethyl ether to aid in the process.
- Quench the organic layer with 15 mL of iced cold DI water and collect the organic layer. Do this three times.
- Neutralize and extract the organic layer with small portions of a saturated sodium bicarbonate solution until the production of CO_2 gas ceases.
- Wash the remaining organic layer with a brine solution then dry the organic over anhydrous sodium sulfate followed by gravity filtration.
- Gently boil off the ether solution in a 50 mL beaker blowing a light stream of air over the top to obtain the crude pentyl valerate.
- Purify the product with simple distillation.
- Analyze the purified product by odor assessment, ATR-IR, ^{1}H NMR (100 MHz), and GC-MS to confirm ester formation and purity.

5 Purification Scheme



6 Results and Data

g of Valeric acid	g of 1-Hexanol	Drops of H_2SO_4	Reflux Time
Mass of Crude Product		Mass of Purified Product	
Theoretical Yield			

Crude Percent Yield	Pure Percent Yield
Crude Product Observations	Purified Product Observations